

अंतरिक्ष रक्षा

ANTARIKSHA- RAKSHA

Space Domain Awareness & Conjunction
Assessment Decision Support

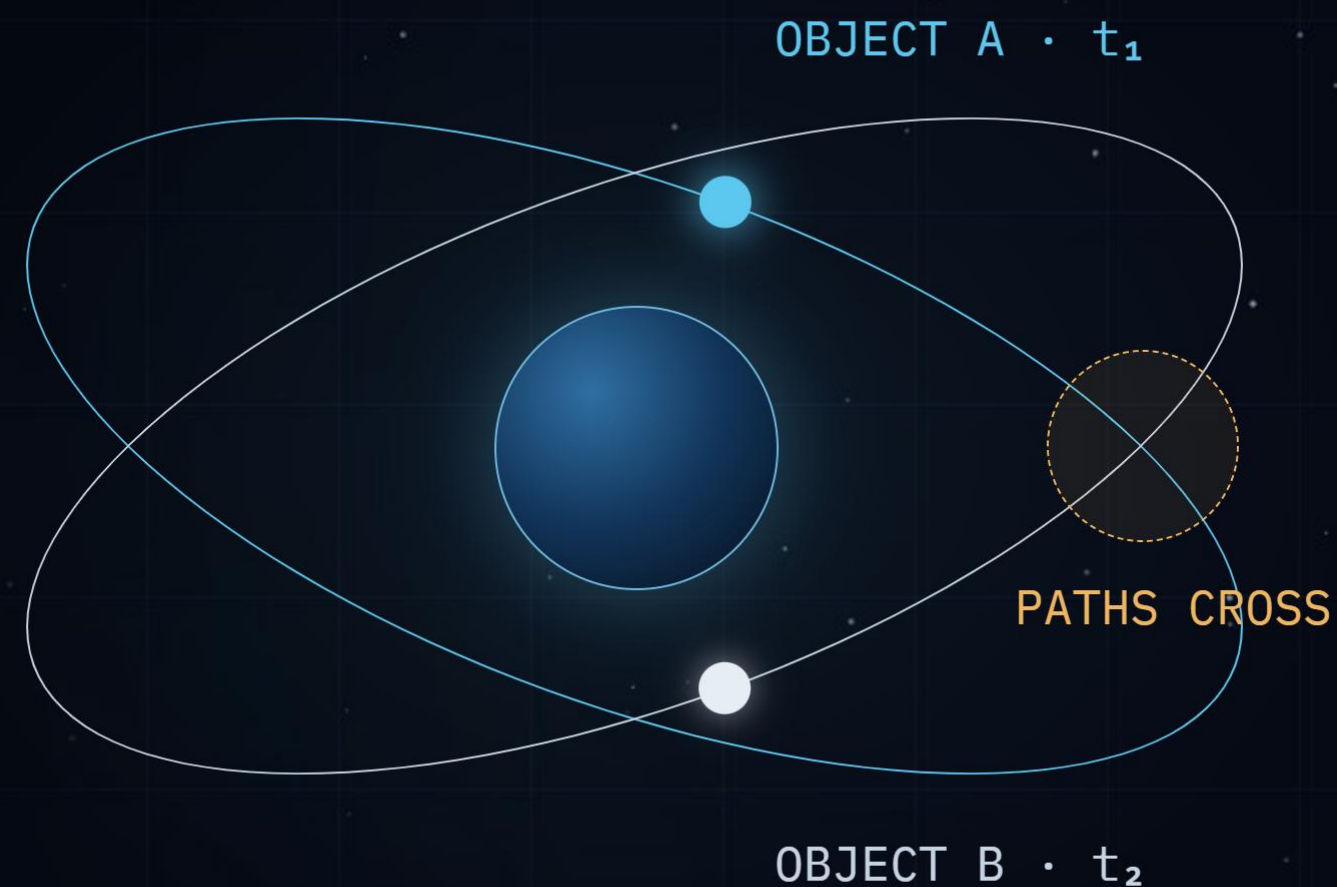
PHYSICS FIRST

AI SECOND

HUMAN DECISION LAST

FUNCTIONAL PROTOTYPE

If two orbital paths cross, do the objects collide?



Orbit is a 3D highway with no visible lanes.

- Satellites and debris move at several km/s.
- Crossing paths alone do not mean collision.

$t_1 \neq t_2$ Different time → safe pass

$t_1 \approx t_2$ Same place + same time → conjunction

Why is tracking alone not enough?

The challenge is turning changing orbital information into a **prioritised, explainable and auditable** workflow.

SPACE IS BECOMING INCREASINGLY ACTIVE

- 01 **More objects, busier orbits**
Satellites and debris share active regimes
- 02 **Information keeps changing**
Every update can shift a prediction
- 03 **Analysts need focus**
Priority, reasons and a record

Why India, why now?

India already has the foundations of Space Situational Awareness. ANTARIKSHA-RAKSHA explores how an explainable, human-gated decision-support layer could help scale the analyst experience as India's space ecosystem and orbital complexity grow.

COMPLEMENTARY TO NATIONAL SSA · NOT A REPLACEMENT

- | | | | |
|----|--|--|--|
| 01 | India's space ecosystem is expanding | | |
| | More missions, more operators, more activity | | |
| 02 | Orbital congestion is increasing | | |
| | Space-traffic complexity keeps growing | | |
| 03 | Space assets matter more | | |
| | To development and to national security | | |
| 04 | Safe operations need capable support | | |
| | Explainable, human-gated decision support | | |

Which questions must the system answer?

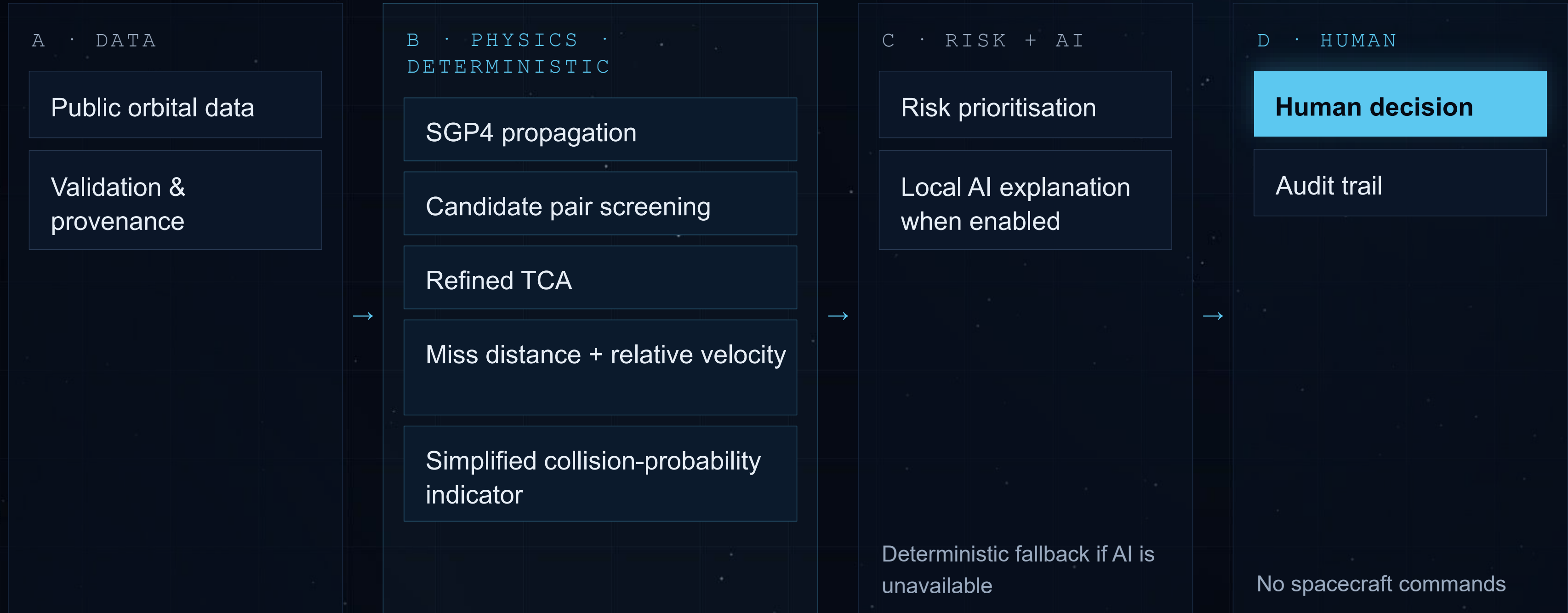
01 SEE	02 PREDICT	03 DETECT	04 UNDERSTAND	05 PRIORITISE	06 EXPLAIN	07 DECIDE
Which objects matter?	When will they come closest?	How close will they be?	What is the estimated risk?	Which event comes first?	How is the evidence explained?	Who makes the final call? A human.

PHYSICS · DETERMINISTIC

RISK + AI

HUMAN

How does orbital data become a decision?



How is each close approach computed?



01	SGP4	Propagates public GP/TLE data
02	Screening	Candidate pairs, configurable horizon
03	TCA	Refined time of closest approach
04	Miss distance	Separation at TCA
05	Relative velocity	Encounter speed and geometry
06	Collision indicator	Analytic, deterministic, no Monte Carlo

QUALIFICATION Simplified analytic collision-probability indicator — NOT an operational covariance-based Pc.

Which event deserves attention first, and why?

01 · EVIDENCE

Deterministic backend output

TCA

MISS DISTANCE

RELATIVE VELOCITY

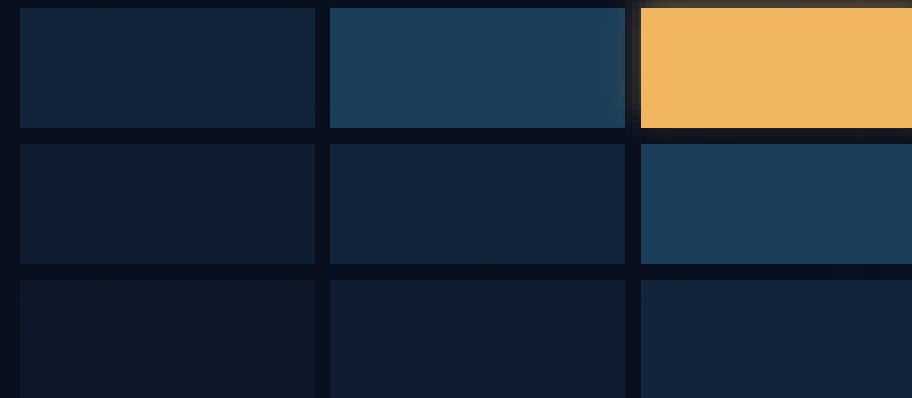
COLLISION INDICATOR

DATA PROVENANCE



02 · PRIORITISATION

Ranked event queue



PROXIMITY →

INDICATOR ↑



Conceptual view

03 · EXPLANATION

Human-readable assessment

Summarises the evidence

States why it was prioritised

Second-pass consistency check
when AI is enabled

AI unavailable? [Deterministic fallback.](#)

Where does AI stop?

AI does

EXPLAIN

- + Summarise deterministic evidence
- + Explain why an event was prioritised
- + Write a human-readable assessment
- + Run a second-pass consistency check when AI is enabled

AI does not

COMPUTE · COMMAND

- × Calculate orbital positions
- × Calculate TCA or miss distance
- × Calculate collision probability
- × Determine orbital mechanics
- × Command spacecraft
- × Make autonomous manoeuvre decisions

Who makes the final decision?



No command is sent to any spacecraft. DESIGN BOUNDARY

TELEGRAM = NOTIFICATION ONLY · ALERTS REACH THE OPERATOR; DECISIONS STAY IN THE CONSOLE

AUTH + RBAC

Viewer

Operator

Asset Manager

Administrator

What runs today?

DEMO SCENARIO — controlled geometry derived from ingested orbital data; not an operational warning.

01 · 3D ORBITAL VIEW

Not an
ING



TIME-LAPSE 60s · 5GPA-PROPAGATED TRACKS · 120 MIN WINDOW

TLE
AGE
SCRI
CANI
COLI
PROJ
RE
ASS
TIER
C
C
C
II
E

LAST
REFI
Perk
AUT
CAD
LAST
ATTI
NEXT
REFI
CURI
TLE
AGE
SCRI
CANI
COLI
PROJ
RE
ASS
TIER
C
C
C
II
E

02 · LIVE DATA + EVENT FEED

EVENT FEED

1 EVENT

◆ DEMO SCENARIO (CONTROLLED SIMULATION)

COLLISION

PROXIMITY

◆ DEMO HISTORY (TWO CONTROLLED SCREENING RUNS)

Step 1 · first observation

Step 2 · next observation

HIGH CONJUNCTION ◆ DEMO

CARTOSAT-3 ↔ FENGYUN 1C

TCA 2026-10-05 22:00 UTC · MISS 0.52 KM · Pc 4.2e-5

LIVE DATA

SOURCE Celestrak public GP/TLE data (celestrak.org)

LAST SUCCESSFUL REFRESH 5 Oct 2026 · 14:00:16 UTC

Periodic public orbital-data refresh

AUTOMATIC REFRESH ENABLED Built-in scheduler

CADENCE EVERY 2 HOURS

LAST ATTEMPT 2026-10-05 14:00:19 UTC

NEXT SCHEDULED REFRESH 2026-10-05 16:00:00 UTC

What changed?

CURRENT INGESTED OBJECTS 103

11 Indian sat · 80 debris · 12 other active sat

TLE/GP EPOCH AGE 0.32-15.4 d (avg 0.86)

SCREENING WINDOW 72 h · STEP 60 s

CANDIDATE PAIRS SCREENED 1012

COLLISION-RISK EVENTS 0

PROXIMITY EVENTS 0

04 · EVIDENCE-BASED BRIEF

4 · DETERMINISTIC BRIEF

○ DETERMINISTIC TEMPLATE (LLM unavailable)

◆ AI UNAVAILABLE — deterministic assessment remains available

AI explains and summarises backend-computed evidence. It does not compute positions, orbits, TCA, miss distance, Pc or risk, does not validate the physics, and cannot command a spacecraft.

CONJUNCTION ASSESSMENT: CARTOSAT-3 (criticality Tier2) has a predicted close approach with FENGYUN 1C (debris) at 2026-10-05T22:00:00.106242+00:00. Predicted miss distance is 0.52 km at a relative velocity of 0.20 km/s, with an estimated collision probability of 4.23e-05 (1 in 23,632) (risk tier: High). Continued tracking until time of closest approach is advised. Illustrative delta-v estimate: ~0.05 m/s (simplified: 1 km separation spread over a 6 h lead time). Not a maneuver plan — does not model orbital geometry, relative velocity, burn location, spacecraft mass, thrust, fuel, or execution constraints. Any maneuver decision rests with the human operator; the delta-v figure is an illustrative estimate, not a maneuver plan.

03 · CONJUNCTION RISK

ASSET

CARTOSAT-3 · Tier2

OBJECT

FENGYUN 1C · Debris (DEMO-ADJUSTED)

RISK TIER

High

TCA (UTC)

2026-10-05 22:00:00 UTC

MISS DISTANCE (AT TCA)

0.520 km

RELATIVE VELOCITY (AT TCA)

0.20 km/s

SEPARATION PROFILE (±45 MIN AROUND TCA)



MAX 183 KM

MIN 0.52 KM @ TCA

COLLISION PROBABILITY (PC)

4.23e-5

≈ 1 IN 23,632

Method: Simplified analytic encounter-plane indicator: the relative position at the refined TCA is projected onto the plane perpendicular to the relative velocity; with independent isotropic per-object sigma, the relative uncertainty in that plane is N(0, 2*sigma^2 I2); Pc = P(|projected relative position| < hard-body radius), evaluated deterministically

Evaluation: deterministic (no sampling)

State evaluated at: TCA (refined)

Position σ: 0.11364869350983797 km per object

Hard-body radius: 20 m

Uncertainty model: Simplified prototype estimate: isotropic Gaussian position sigma per object, growing linearly with time-to-TCA. NOT full covariance propagation.

05 · HUMAN DECISION + AUDIT

Physics evidence is generated by the deterministic backend. AI explains that evidence.

AI DID (this event)

Nothing for this event — the brief is a deterministic template (local LLM unavailable)

AI DID NOT

Calculate orbits or positions

Calculate the TCA

Calculate the miss distance

Calculate the Pc

Determine physical collision risk or the risk tier

Command any spacecraft

Plan or execute maneuvers

Make the final decision — the operator does

5 · HUMAN DECISION

APPROVE ASSESSMENT

DISMISS

Records an operator decision in the audit trail. No command is sent to any spacecraft.

Feedback: In-context feedback: up to 3 recent operator rejection reasons are added to the prompt; no model weights are modified

AUDIT HISTORY

No decisions recorded for this event yet.

■ Vercel frontend ■ Railway backend ■ Persistent database ■ 2-hour scheduler ■ Auth + RBAC ■ Public data refresh

Where does it fit in India's space capability?

India already has significant space surveillance and tracking capabilities. ANTARIKSHA-RAKSHA is a **complementary software-layer prototype** that turns orbital evidence into prioritised, explainable and auditable decision support.

MODULAR

Software layer that can sit alongside existing systems

OPTIONAL LOCAL AI

Explanations from a local model when enabled, with deterministic fallback

GOVERNED

Human-in-the-loop decisions with a persistent audit trail

EXTENSIBLE

Scalable architecture, ready for authoritative data sources

NASA, ESA and commercial SDA providers already run sophisticated conjunction assessment. Our contribution is the integrated workflow.

EVIDENCE → PRIORITY → LOCAL AI → HUMAN → AUDIT

What could this enable?

Potential future value, **not implemented capability**: what an explainable, human-gated decision-support layer could enable as orbital activity grows.

01 · PRIORITISE

Focus analyst attention on the most important conjunctions

02 · EXPLAIN

Turn complex orbital evidence into concise, traceable assessments

03 · SCALE

Support analysts as object populations and event volumes grow

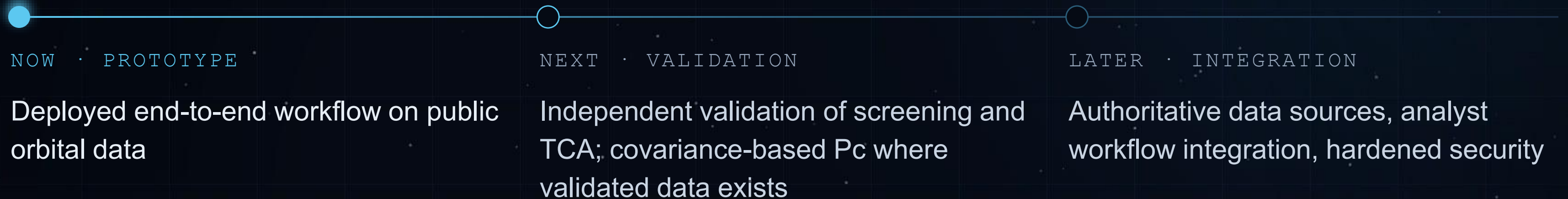
04 · GOVERN

Keep AI bounded by deterministic evidence and human approval

Future foundation: authoritative ephemerides + covariance / CDM integration + institutional analyst workflows.

FUTURE FOUNDATION · POTENTIAL, NOT BUILT

What are the limits, and what comes next?



PROTOTYPE LIMITATIONS

Current live ingestion uses public CelesTrak GP/TLE data — accuracy bounded by source

Demo scenario uses controlled geometry

Not a command, autonomous-avoidance or weapon system

Simplified indicator, not operational covariance-based Pc

Not a replacement for ISRO / IS4OM / NETRA / DRDO systems

Pathway above is potential, not committed

— अंतरिक्ष रक्षा

Physics first. AI second. Human decision last.

From orbital data to explainable decision support.

DATA → PHYSICS → RISK → AI EXPLANATION → HUMAN DECISION → AUDIT